

One new move, a new world

Name: _____

Date: _____

Use TWO piles. You may remove any positive number from one pile, OR remove the SAME positive number from both. Take the last counter to win. This is Wythoff's game.

1. Break the matching strategy.

Why is $(4, 4)$ no longer a trap? Classify all starts (a, b) with $0 \leq a \leq b \leq 7$. Work in order of $a + b$; a trap has no move to an earlier trap. Treat $(0, 0)$ as terminal.

2. Build a smaller list.

Start with $(0, 0)$. Repeatedly choose the smallest nonnegative integer not yet in either column as a_n , then put $b_n = a_n + n$.

n	0	1	2	3	4	5
a_n	0	1				
b_n	0	2				

Does this list match your small-game traps? Explain why no rook move and no equal-take move can connect two listed pairs, even when pile order is swapped.

What is proved? What still needs to be proved?

3. An unexpected number.

Let $\phi = (1 + \sqrt{5})/2 \approx 1.618$. A known theorem says the pairs are $(\lfloor n\phi \rfloor, \lfloor n\phi^2 \rfloor)$, together with swapped pairs. Floors mean round down. Test $n = 1, \dots, 5$. Why does $\phi^2 = \phi + 1$ explain the differences $b_n - a_n = n$?

Harder: To prove the list is the whole losing set, what must you show about a position *outside* it? Try finding such moves from $(2, 4)$, $(5, 8)$, and $(8, 12)$.

Try it with objects. Explain by pointing, talking, or drawing.